

Dynamic curvature-adaptive MPCC with physical feasibility constraints for high-speed racing under frequent curvature variations

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ABSTRACT

In high-speed racing environments, frequent changes in track curvature can compromise the stability of control systems, thereby limiting the racing performance of autonomous vehicles. To address this challenge, an enhanced model predictive contouring control algorithm integrating a curvature-aware dynamic error adaptation mechanism with physical feasibility constraints (CA-PF-MPCC) is proposed. The algorithm employs a dynamic weight adjustment function based on trajectory curvature, adaptively balancing contouring and lag errors to enhance responsiveness in sections with varying geometric characteristics. Additionally, by combining the friction ellipse theory with curvature-velocity coupling constraints, a unified control-speed feasible domain is established, which effectively suppresses physical instability during aggressive maneuvers. Experimental results indicate that, under scenarios with frequent curvature changes, CA-PF-MPCC reduces racing time by 3.263 s, 2.715 s, 4.289 s, and 1.720 s compared to the Feedforward-Feedback, MPCC, MPC-CBF, and LPV-MPC algorithms, respectively. The rate of change of the heading angle decreases by 0.062 rad/s, 0.070 rad/s, 0.139 rad/s, and 0.038 rad/s, correspondingly, thereby effectively suppressing heading fluctuations and improving attitude stability. These findings demonstrate that the proposed method significantly enhances racing performance while improving dynamic adaptability and high-speed stability on complex tracks, highlighting its practical relevance.

1. Introduction

With the rapid advancement of artificial intelligence and intelligent driving technologies, autonomous racing has emerged as a state-of-the-art testbed that integrates perception, decision-making, and control, enabling the evaluation of autonomous driving systems under extreme conditions. Recent international competitions, including Roborace, the Indy Autonomous Challenge (IAC), F1Tenth, and the Abu Dhabi Autonomous Racing League (A2RL), have significantly accelerated the development of intelligent racing technologies in high-speed, complex, and dynamic environments. In these scenarios, autonomous racing vehicles are required to operate near physical handling limits while maintaining stability and performance, thereby imposing stringent requirements on modeling accuracy and control robustness. In such high-speed autonomous racing scenarios, track curvature often exhibits frequent and abrupt variations, which poses a fundamental challenge to trajectory tracking control. Specifically, rapid curvature changes make the intrinsic trade-off between tracking accuracy and longitudinal progress difficult to adjust online, leading to a pronounced degradation in control

performance. Moreover, under high-speed operating conditions, physical feasibility constraints such as tire force saturation and steering rate limits are frequently activated, substantially shrinking the feasible control input space and further undermining the stability and robustness of the closed-loop system.

1.1. Related works

Vehicle dynamics modeling is crucial for achieving high-performance trajectory tracking and precise control in autonomous racing cars, serving as a fundamental component of the overall system. Based on modeling accuracy, common approaches are typically classified into two categories: kinematic models and dynamic models (Zhang et al., 2024). References (Alrifaae & Maczjewski, 2018; Subosits & Gerdes, 2019) simplify the vehicle as a point mass at its center of mass and model its longitudinal and lateral motions in a two-dimensional plane using Newton's second law; however, these approaches do not fully account for the influence of lateral forces induced by posture changes on tire grip and overall vehicle performance. To balance

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modeling accuracy and computational efficiency, Jain et al. (2021) proposed a simplified bicycle model to capture vehicle dynamics and combined it with the Pacejka tire model to simulate nonlinear tire forces, thereby enhancing modeling fidelity. Furthermore, Subosits and Gerdes (2021) employed a dynamic double-track model along with the Fiala tire model to more accurately represent the longitudinal and lateral forces at each wheel, improving the predictive capability of the vehicle's local dynamic response. To increase the model's adaptability to real-world conditions, Wachter et al. (2020) incorporated aerodynamic effects into dynamic modeling. While this improves physical consistency, it also imposes higher parameter requirements and computational overhead, limiting the model's real-time applicability.

Designing an efficient control strategy based on accurate vehicle modeling is essential for ensuring the stable operation of autonomous racing cars in high-speed and complex environments. Kapania and Gerdes (2015) proposed a feedforward-feedback control method that suppresses lateral tracking errors to a certain extent. However, the method employs fixed control parameters, which makes it difficult to adapt to dynamic environments with drastic track curvature changes, thereby reducing control performance. In contrast, Model Predictive Control (MPC) has gained considerable attention in recent years owing to its strong constraint-handling capabilities and robustness. For example, Vázquez et al. (2020) developed a hierarchical motion control framework in which the upper layer computes the optimal racing trajectory using an optimization algorithm, while the lower layer employs MPC for trajectory tracking, with performance validated on a real track. Guo et al. (2024) incorporated friction-ellipse constraints into an LTV-MPC framework to jointly restrict traction and lateral force limits, and adopted gradual linearization to improve real-time performance. Zeng et al. (2021) integrated MPC with a discrete-time control barrier function to enhance safety constraints and theoretically analyzed the algorithm's stability and feasibility. Similarly, Kalaria et al. (2024) embedded an adaptive delay compensator into Tube-MPC to mitigate latency during high-speed driving, thereby enhancing control responsiveness. Zarrouki et al. (2024) further improved robustness by integrating the uncertainty propagation horizon and polynomial chaos expansion into a stochastic nonlinear MPC framework, effectively increasing adaptability to modeling errors and disturbances. However, many of these approaches primarily emphasize safety or robustness, and do not explicitly address performance degradation caused by rapidly varying track curvature in racing scenarios.

Recent years have witnessed increasing interest in Model Predictive Contouring Control (MPCC), an important variant of MPC, owing to its superior performance in simultaneous path tracking and speed regulation for unmanned aerial vehicles and autonomous racing platforms (Ji et al., 2020; Romero et al., 2022). Liniger et al. (2015) applied MPCC to scale model racing control, demonstrating good real-time and tracking performance. Li et al. (2024) introduced track curvature information for autonomous racing path planning, improving racing performance. However, most existing MPCC formulations rely on fixed cost weights, which limits their adaptability to rapidly varying curvature and often leads to degraded stability or conservative behavior in high-speed racing scenarios.

In parallel, data-driven approaches—particularly deep reinforcement learning (DRL)—have emerged as promising alternatives for adaptive decision-making and control. DRL enables autonomous acquisition of near-optimal control strategies through interaction with the environment without relying on explicit system models. For example, Hell et al. (2024) proposed a model-free DRL framework combined with LiDAR-based perception, achieving improved dynamic response accuracy in simulation. However, high-speed operation led to instability phenomena such as excessive attitude oscillations. Evans et al. (2023) developed a trajectory-aided DRL approach with an end-to-end architecture, which improved adaptability to complex environments and demonstrated favorable speed-stability trade-offs in simulation. Nevertheless, DRL-based methods face persistent obstacles—including low sample ef-

ficiency, unstable convergence, and high sensitivity to disturbances—which currently confine their applicability to simulation environments rather than real-world deployment.

In summary, although both model-based and learning-based control strategies have achieved notable progress, maintaining stability and performance under high-speed driving with frequent curvature variations remains an open challenge. This motivates the development of control frameworks that can explicitly adapt to track curvature while rigorously respecting vehicle physical feasibility.

1.2. Contribution and structure

To overcome the aforementioned limitations, this study proposes a Curvature-Aware and Physical-Feasibility-Enhanced Model Predictive Contouring Control (CA-PF-MPCC) algorithm, which systematically enhances racing performance on tracks with frequently varying curvature. Specifically, the main contributions of this work are as follows:

- **Curvature-Aware Dynamic Error Adaptation:** A curvature-dependent weight adjustment function is developed to dynamically regulate the cost coefficients of contouring and lag errors within the MPCC framework. This enables adaptive responses to road geometry variations and improves lap-time performance under high-speed conditions with frequent curvature changes.
- **Integration of Physical Feasibility Constraints:** A friction-ellipse-based tire force constraint is incorporated in conjunction with a curvature-velocity coupling constraints to construct a unified feasible domain for velocity and control inputs. This formulation mitigates physical instability under extreme operating conditions and increases the control safety margin.
- **System Validation under High-speed Conditions:** A high-fidelity autonomous racing simulation platform is employed to comprehensively evaluate the proposed algorithm's racing performance in high-speed scenarios, demonstrating its potential for practical deployment in real-world racing tasks.

The rest of this article is organized as follows. Section 2 describes the development of the autonomous racing vehicle model and outlines the baseline control algorithm. Section 3 details the proposed methodology in detail. Section 4 presents experimental results and comparative analyses with existing methods. Finally, Section 5 draws conclusions.

2. Autonomous racing vehicle modeling and baseline control algorithm

2.1. Vehicle dynamics modeling

To satisfy the requirements of real-time computation and precise control in high-speed autonomous racing, this study employs the widely adopted bicycle dynamics model (Alcalá et al., 2020; Dai et al., 2023; Liniger et al., 2015), as illustrated in Fig. 1. This model provides an effective balance between computational efficiency and physical fidelity, making it suitable for high-speed racing scenarios where aggressive maneuvers and rapid state variations must be handled in real time. The model is constructed under the following simplifying assumptions:

- (1) Vertical motion along the Z-axis is neglected;
- (2) The left and right tires are assumed to exhibit identical steering angles and wheel speeds at all times;
- (3) The vehicle mass is concentrated at the center of gravity. Within this framework, the vehicle's lateral dynamics under high-speed racing conditions are primarily governed by nonlinear lateral tire forces, which are strongly dependent on the sideslip angle and play a critical role in determining cornering stability and performance limits.

Let $\mathbf{x} = [x, y, \varphi]^T$ denote the vehicle's position in the inertial coordinate system, where x and y represent the horizontal and vertical coordinates of the vehicle's center of mass, respectively, and φ denotes

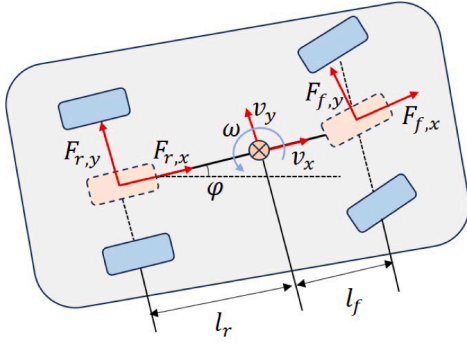


Fig. 1. Dynamic model of the autonomous racing vehicle.

the heading angle of the longitudinal axis relative to the inertial X -axis. The state vector in the body-fixed coordinate system is defined as $\mathbf{v} = [v_x, v_y, \omega]^T$, where v_x and v_y represent the longitudinal and lateral velocities, respectively, and ω is the vehicle's angular velocity. These state variables are particularly important for high-speed autonomous racing, as lateral velocity and yaw dynamics directly reflect vehicle stability margins under aggressive cornering. Based on these state definitions and modeling assumptions, the vehicle dynamics can be expressed as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \\ \dot{v}_x \\ \dot{v}_y \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v_x \cos(\varphi) - v_y \sin(\varphi) \\ v_x \sin(\varphi) + v_y \cos(\varphi) \\ \omega \\ \frac{1}{m} (-F_{f,y} \sin \delta + F_{r,x} + F_{f,x} \cos \delta + m v_y \omega) \\ \frac{1}{m} (F_{r,y} + F_{f,y} \cos \delta + F_{f,x} \sin \delta - m v_x \omega) \\ \frac{1}{I_z} (l_f F_{f,y} \cos \delta + l_f F_{f,x} \sin \delta - l_r F_{r,y}) \end{bmatrix}, \quad (1)$$

where m is the vehicle mass, I_z is the yaw moment of inertia, l_f and l_r are the distances from the center of mass to the front and rear axles, respectively, δ is the steering angle, and $F_{f,y}$, $F_{r,y}$, $F_{f,x}$, and $F_{r,x}$ denote the front and rear lateral forces and the front and rear longitudinal forces, respectively. Since the racing vehicle considered in this study is powered by an internal combustion engine, adopts a rear-wheel-drive and front-wheel-steering configuration, and is equipped with independently actuated braking systems on all four wheels, the longitudinal tire forces $F_{f,x}$ and $F_{r,x}$ are approximated using engine and braking torque modeling. These forces are formulated as:

$$F_{f,y} = D_f \sin(C_f \arctan(B_f \alpha_f)), \quad (2)$$

$$F_{r,y} = D_r \sin(C_r \arctan(B_r \alpha_r)), \quad (3)$$

$$F_{r,x} = \frac{T_{\text{engine}} - T_{\text{brake}}}{R_w} - C_{r1} - C_{r2} v_x^2, \quad (4)$$

$$F_{f,x} = -\frac{T_{\text{brake}}}{R_w} \quad (5)$$

where the engine driving torque T_{engine} is determined by the throttle input and engine speed, and its relationship is obtained by fitting engine dynamometer data. The braking torque T_{brake} is governed by the brake pedal input and modeled according to the braking system characteristics. C_{r1} denotes the rolling resistance coefficient, C_{r2} represents the aerodynamic drag coefficient, and R_w is the effective tire radius. Eqs. (2) and (3) are derived from Pacejka's Magic Formula to capture nonlinear tire behavior and saturation effects that are critical in high-speed racing. The front and rear tire slip angles, α_f and α_r , are given by:

$$\alpha_f = \delta - \arctan\left(\frac{\omega l_f + v_y}{v_x}\right), \quad (6)$$

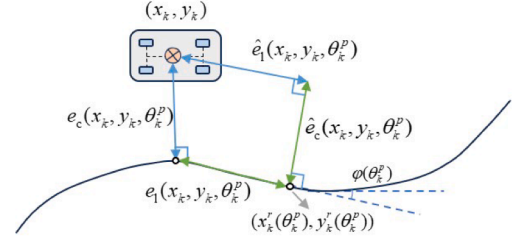


Fig. 2. Geometric relationship between the contouring and lag errors.

$$\alpha_r = \arctan\left(\frac{\omega l_r - v_y}{v_x}\right), \quad (7)$$

where δ denotes the steering angle of the front wheels. This bicycle model provides an effective balance between computational efficiency and physical fidelity for high-speed autonomous racing, while not explicitly capturing extreme-limit effects such as aerodynamic load variations, suspension dynamics, or combined-slip tire behavior.

2.2. Model predictive contouring control framework

Building on the established vehicle dynamics model, this study adopts the model predictive contouring control (MPCC) algorithm to achieve high-performance and robust trajectory tracking for autonomous racing vehicles. MPCC, which integrates the principles of model predictive control with direct optimization of path contour errors, has been widely applied in scenarios such as UAV racing (Ji et al., 2020; Krinner et al., 2024), high-speed autonomous driving (Picotti et al., 2023), and coordinated formation control (Zhao & Li, 2024). Unlike conventional path-tracking methods, MPCC considers not only the lateral deviation from the reference trajectory but also the lag error, which constrains the vehicle's longitudinal position along the path, thereby enhancing tracking accuracy and stability. By performing real-time optimization of the vehicle's predicted trajectory, MPCC can effectively cope with the complex dynamic environments encountered during high-speed operation, fulfilling the requirements for precise path following in racing scenarios.

Within the MPCC framework, the vehicle's deviation from the reference trajectory is typically decomposed into contouring error (e_c) and lag error (e_l), as illustrated in Fig. 2. The contouring error e_c quantifies the vehicle's lateral displacement perpendicular to the reference path, reflecting lateral tracking performance, whereas the lag error e_l measures the displacement along the tangent of the path, indicating longitudinal advance or delay relative to the reference trajectory. These errors are computed as:

$$e_c(x_k, y_k, \theta_k^p) = \sin(\varphi(\theta_k^p))(x_k - x_k^r(\theta_k^p)) - \cos(\varphi(\theta_k^p))(y_k - y_k^r(\theta_k^p)) \quad (8)$$

$$e_l(x_k, y_k, \theta_k^p) = -\cos(\varphi(\theta_k^p))(x_k - x_k^r(\theta_k^p)) - \sin(\varphi(\theta_k^p))(y_k - y_k^r(\theta_k^p)) \quad (9)$$

where (x_k, y_k) denotes the current position of the autonomous racing vehicle, $(x_k^r(\theta_k^p), y_k^r(\theta_k^p))$ represents the desired position on the reference trajectory parameterized by arc length using cubic splines, and $\varphi(\theta_k^p) = \arctan\left(\frac{\partial y_k^r(\theta_k^p)}{\partial x_k^r(\theta_k^p)}\right)$ is the tangent angle of the reference path at the corresponding point relative to the X -axis.

3. Construction of the CA-PF-MPCC control framework

Building on the established vehicle dynamics model, this study develops an enhanced model predictive contouring control algorithm specifically tailored for high-speed autonomous racing on tracks characterized by frequent and abrupt curvature variations. In such scenarios,

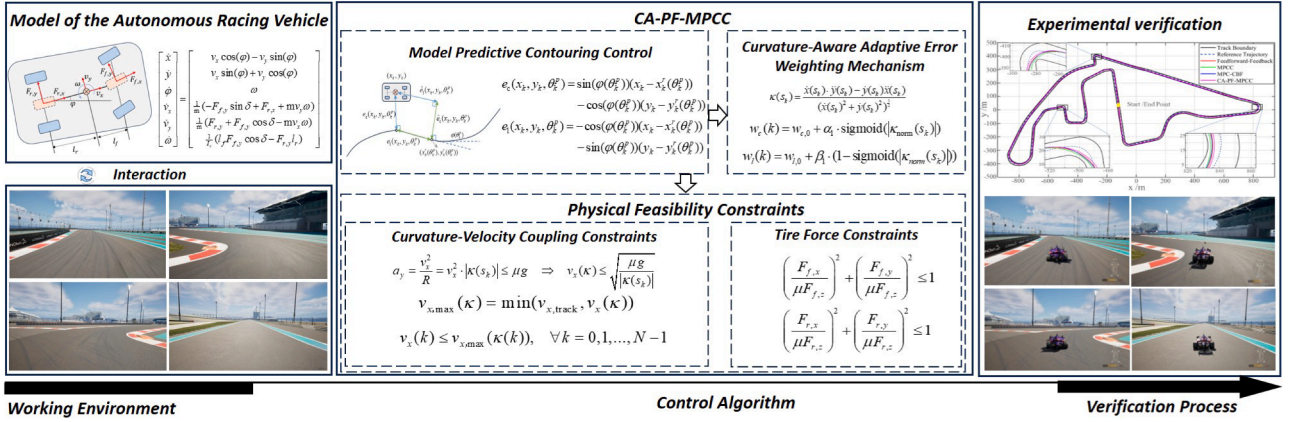


Fig. 3. Overall framework of the proposed CA-PF-MPCC algorithm for autonomous racing.

conventional MPCC formulations with fixed error weights often struggle to simultaneously maintain lateral tracking accuracy in sharp corners and sufficient longitudinal progress along low-curvature segments, which may result in suboptimal lap times or degraded vehicle stability. To address these challenges, the proposed algorithm introduces a curvature-aware adaptive error weighting mechanism and explicitly incorporates physical feasibility constraints into the MPCC optimization problem, thereby enhancing both racing performance and operational safety under aggressive high-speed driving conditions. The overall architecture of the proposed algorithm is illustrated in Fig. 3.

3.1. Curvature-aware adaptive error weighting mechanism

Under high-speed or extreme operating conditions, conventional MPCC controllers with fixed cost function weights are often inadequate to meet the diverse performance demands of autonomous racing vehicles across varying scenarios. This limitation is particularly pronounced when navigating track segments with rapidly changing curvature, where system stability and trajectory tracking accuracy can deteriorate, ultimately constraining overall racing performance. To address this issue and enhance both adaptability to track geometry and control accuracy, a curvature-aware adaptive error weighting mechanism is proposed within the trajectory tracking cost function. Specifically, when the curvature κ of the reference trajectory is large (e.g., sharp turns), higher lateral accuracy is required. In such cases, the contouring error weight w_c is increased to ensure precise trajectory following and improved cornering stability. Conversely, when $\kappa \approx 0$ (e.g., straight segments or gentle curves), propulsion efficiency becomes the priority. Accordingly, the lag error weight w_l is increased to strengthen forward-tracking capability along the tangential direction. Based on these principles, the adaptive weighting scheme is defined as:

$$w_c(k) = w_{c,0} + \alpha_1 \cdot \text{sigmoid}(|\kappa_{\text{norm}}(s_k)|), \quad (10)$$

$$w_l(k) = w_{l,0} + \beta_1 \cdot (1 - \text{sigmoid}(|\kappa_{\text{norm}}(s_k)|)), \quad (11)$$

where $w_{c,0}$ and $w_{l,0}$ denote the nominal weights of contouring and lag errors, respectively; α_1 and β_1 are hyperparameters regulating the influence of curvature; $\kappa_{\text{norm}}(s_k)$ denotes the normalized curvature of the reference trajectory at path progress s_k , whose value is scaled to the range $[0, 1]$; and $\text{sigmoid}(\cdot)$ is a continuously differentiable function that ensures smooth variation of the weights, with its slope denoted by α_0 . The curvature $\kappa(s_k)$ of the reference trajectory at prediction step k is calculated as:

$$\kappa(s_k) = \frac{\dot{x}(s_k)\ddot{y}(s_k) - \dot{y}(s_k)\ddot{x}(s_k)}{(\dot{x}(s_k)^2 + \dot{y}(s_k)^2)^{3/2}}, \quad (12)$$

where $\dot{x}(s_k)$ and $\ddot{x}(s_k)$ denote the first- and second-order discrete derivatives of the reference trajectory in the x -direction, respectively. Specifically, the first-order derivative is computed using a forward difference

approximation as $\dot{x}(s_k) = x_{s,k} - x_{s,k-1}$, while the second-order derivative is obtained by differencing consecutive first-order derivatives, i.e., $\ddot{x}(s_k) = \dot{x}(s_k) - \dot{x}(s_{k-1})$. Here, $x_{s,k}$ and $x_{s,k-1}$ represent the reference trajectory positions in the x -direction at prediction steps k and $k-1$, respectively. The derivatives $\dot{y}(s_k)$ and $\ddot{y}(s_k)$ are computed in the same manner for the y -direction. Consequently, the trajectory-tracking cost function can be formulated as

$$J_{\text{track}} = \sum_{k=1}^{N-1} (w_c(k) \cdot e_{c,k}^2 + w_l(k) \cdot e_{l,k}^2), \quad (13)$$

The incorporation of the curvature-aware adaptive error-weighting mechanism significantly enhances the controller's ability to dynamically adjust error penalties across varying track segments, thereby improving both racing performance and system stability, especially under high-speed conditions with frequent curvature changes.

3.2. Tire force constraints

Building on the curvature-aware adaptive error weighting mechanism, this study further enhances the controller's capability to respect the vehicle's dynamic limits by incorporating tire force constraints into the MPCC optimization problem. These constraints explicitly restrict the feasible solution space of the nonlinear optimization problem to physically admissible tire force combinations, thereby improving the physical feasibility and robustness of the control system. Using a friction ellipsoid model, the constraints restrict the total force applied by each tire in longitudinal traction and lateral steering, ensuring it does not exceed the maximum allowable limit determined by tire-road adhesion. Specifically, the following nonlinear inequality constraints are imposed on the front and rear wheels:

$$\left(\frac{F_{f,x}}{\mu F_{f,z}} \right)^2 + \left(\frac{F_{f,y}}{\mu F_{f,z}} \right)^2 \leq 1, \quad (14)$$

$$\left(\frac{F_{r,x}}{\mu F_{r,z}} \right)^2 + \left(\frac{F_{r,y}}{\mu F_{r,z}} \right)^2 \leq 1, \quad (15)$$

where μ denotes the friction coefficient between the tire and the ground, and $F_{f,z}$ and $F_{r,z}$ are the normal loads on the front and rear wheels, respectively. Each force component is computed using the vehicle dynamics model in conjunction with the magic formula tire model. These constraints prevent the optimizer from selecting control actions that violate fundamental tire force limits, thereby avoiding wheel slip, lateral sliding, and loss of control due to commands exceeding the tires' physical limits, making them particularly suitable for high-speed or aggressive maneuvering of autonomous racing vehicles.

3.3. Curvature-velocity coupling constraints

Building upon the previously established tire force constraints, this study introduces a curvature-velocity coupling constraint to regulate the speed of the autonomous racing car in high-curvature sections, thereby improving the overall stability of the MPCC algorithm. where a_y is the lateral acceleration, R is the radius of curvature of the path (the reciprocal of which is $\kappa(s_k)$), μ denotes the friction coefficient between the tires and the ground, and g is the gravitational acceleration. To improve numerical stability, the nonlinear velocity upper bound is expressed as

$$v_{x,\max}(\kappa) = \min(v_{x,\text{track}}, v_x(\kappa)), \quad (17)$$

where $v_{x,\text{track}}$ represents the reference velocity generated by the trajectory planning algorithm. This constraint is enforced as a hard limit during the MPCC control cycle:

$$v_x(k) \leq v_{x,\max}(\kappa(k)), \quad \forall k = 0, 1, \dots, N-1. \quad (18)$$

Notably, this curvature-velocity coupling becomes active primarily in high-curvature regions where lateral acceleration demand approaches tire force limits, while remaining inactive on straight or low-curvature segments. By imposing this curvature-velocity coupling, the controller prevents overspeeding in tight curves, mitigating the risk of rollovers, while allowing the vehicle to exploit its maximum speed on straight segments. This constraint thus improves both stability and competitive performance.

3.4. Formulation of the improved MPCC optimization problem

Finally, by integrating the curvature-aware adaptive error-weighting mechanism, tire force constraints, and curvature-velocity coupling constraints into the MPCC framework, the resulting nonlinear optimization problem can be formulated as:

$$\min_{\mathbf{u}_0: N-1, \xi_1: N} \sum_{k=0}^{N-1} \left(w_c(k) \cdot e_{c,k}^2 + w_l(k) \cdot e_{l,k}^2 \right) + \sum_{k=0}^{N-1} \|\Delta \mathbf{u}_k\|_{R_1}^2, \quad (19a)$$

$$\text{s.t. } \xi_{k+1} = f(\xi_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \quad (19b)$$

$$\xi_0 = \xi_{\text{current}}, \quad (19c)$$

$$\Delta \mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}, \quad (19d)$$

$$\mathbf{u}_{\min} \leq \mathbf{u}_k \leq \mathbf{u}_{\max}, \quad (19e)$$

$$\Delta \mathbf{u}_{\min} \leq \Delta \mathbf{u}_k \leq \Delta \mathbf{u}_{\max}, \quad (19f)$$

$$\left(\frac{F_x}{\mu F_z} \right)^2 + \left(\frac{F_y}{\mu F_z} \right)^2 \leq 1, \quad (19g)$$

$$v_x(k) \leq v_{x,\max}(\kappa(k)), \quad (19h)$$

Where $\xi_k \in \mathbb{R}^n$ and $\mathbf{u}_k \in \mathbb{R}^m$ denote the system state and control input at step k , respectively. In the present formulation, the state vector at step k is defined as $\xi_k = [x_k, y_k, \varphi_k]^T$, where x_k and y_k are the planar coordinates and φ_k is the yaw angle. Similarly, the control vector at step k is defined as $\mathbf{u}_k = [v_{x,k}, v_{y,k}, \omega_k]^T$, consisting of the longitudinal velocity $v_{x,k}$, lateral velocity $v_{y,k}$, and yaw rate ω_k . $f(\cdot)$ represents the nonlinear vehicle dynamics, $e_{c,k}$ and $e_{l,k}$ are the contouring and lag errors, and $w_c(k)$ and $w_l(k)$ are curvature-aware dynamic weighting functions that adaptively modulate the relative significance of these errors based on the local track geometry. R_1 is the weighting matrix for the control input rate. Constraint (19g) enforces tire force limits to ensure physical feasibility and prevent the vehicle from exceeding tire friction capacity during high-speed or high-curvature maneuvers, while constraint (19h) imposes a curvature-velocity coupling to limit vehicle speed in sharp turns, thereby enhancing the physical consistency of the MPCC controller. Infeasibility may arise when the reference trajectory or commanded velocity violates fundamental physical limits of the vehicle, reflecting physically unrealizable demands rather than numerical artifacts of the optimization. Collectively, this optimization framework enables

the controller to achieve adaptive trajectory tracking while preserving vehicle stability under high-speed conditions with dynamically varying curvature, thereby significantly improving the racing performance of autonomous vehicles. The overall CA-PF-MPCC procedure is summarized in Algorithm 1.

Algorithm 1 CA-PF-MPCC.

Require: Current vehicle state (x_k, y_k, φ_k) ; local reference trajectory $(x_{\text{ref}}, y_{\text{ref}}, \varphi_{\text{ref}})$

Ensure: Optimal control input at the current time step u_0^*

- 1: **Initialization:** prediction horizon N , sampling time Δt , cost weights, vehicle model parameters; configure the IPOPT solver
 - 2: **while** the vehicle is operating **do**
 - 3: Generate reference trajectory within the prediction horizon using cubic spline interpolation: $(x_k^r(\theta_k^p), y_k^r(\theta_k^p), \varphi_k^r(\theta_k^p))$
 - 4: Compute curvature using Eq. (12), and normalize it to obtain $\kappa_{\text{norm}}(s_k)$
 - 5: Define MPC decision variables: predicted state sequence $\{x_k\}_{k=0}^N$ and control input sequence $\{u_k\}_{k=0}^{N-1}$
 - 6: **for** $k \leftarrow 0$ **to** $N-1$ **do**
 - 7: $w_c(k) \leftarrow w_{c,0} + \alpha_1 \cdot \text{sigmoid}(|\kappa_{\text{norm}}(s_k)|)$
 - 8: $w_l(k) \leftarrow w_{l,0} + \beta_1 \cdot (1 - \text{sigmoid}(|\kappa_{\text{norm}}(s_k)|))$
 - 9: Compute contouring error $e_{c,k}$ and lag error $e_{l,k}$
 - 10: **end for**
 - 11: Construct nonlinear optimization constraints according to Eq. (19b)–(19h)
 - 12: Construct the objective function according to Eq. (19a)
 - 13: Solve the nonlinear programming problem using IPOPT with warm-start initialization
 - 14: Obtain the optimal control sequence $\{u_k^*\}_{k=0}^{N-1}$
 - 15: Apply the first control input u_0^* to the vehicle
 - 16: Update the vehicle state and shift the prediction horizon forward by one time step
 - 17: **end while**
-

3.5. Closed-loop stability discussion

Theorem 1 (Conditional Local Stability of CA-PF-MPCC). *Consider the discrete-time vehicle dynamics*

$$\xi_{k+1} = f(\xi_k, \mathbf{u}_k), \quad (20)$$

where $\xi_k \in \mathbb{R}^n$ denotes the system state and $\mathbf{u}_k \in \mathbb{R}^m$ denotes the control input at time step k . The control input is generated by the proposed CA-PF-MPCC scheme, which solves a receding-horizon optimization problem incorporating curvature-aware adaptive error weighting, tire force constraints, curvature-velocity coupling constraints, and input rate constraints.

Assume that (i) the CA-PF-MPCC optimization problem is feasible at the initial state ξ_0 ; (ii) the vehicle dynamics $f(\xi, \mathbf{u})$ are locally Lipschitz continuous in a neighborhood of the reference trajectory; (iii) the stage cost is positive definite with respect to the contouring and lag errors as well as the control input deviations; (iv) the adaptive error weighting remains bounded, continuous, and varies slowly; and (v) the additional constraints introduced by CA-PF-MPCC are either inactive in a sufficiently small neighborhood of the reference trajectory, or can be approximated by locally valid convex inner sets. Under these conditions, recursive feasibility of the receding-horizon optimization problem is maintained. Then, under the above conditions, the closed-loop system under CA-PF-MPCC is locally asymptotically stable with respect to the reference trajectory.

Proof. The closed-loop stability of linear time-varying model predictive contouring control has been rigorously established under standard contractive MPC assumptions (Lam et al., 2010), providing a theoretical reference for the following discussion. In CA-PF-MPCC, the receding-horizon formulation is maintained, while the stage cost and admissible control set are modified to incorporate curvature-aware weighting and

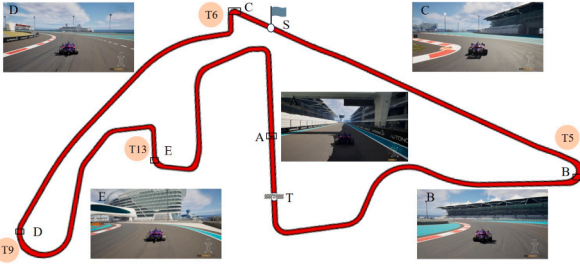


Fig. 4. Layout of the autonomous racing track.

physical feasibility constraints. Since the adaptive weighting is strictly positive and bounded, local positive definiteness of the stage cost is preserved. Moreover, under the stated assumptions on constraint activity and convex inner approximations, recursive feasibility can be maintained.

Define the Lyapunov candidate function as the optimal cost-to-go of the CA-PF-MPCC problem at time step k . By applying the standard horizon-shifting argument used in contractive MPCC, a feasible candidate solution at time step $k+1$ can be constructed from the optimal solution at time step k . Consequently, the Lyapunov difference along the closed-loop trajectory satisfies

$$V(\xi_{k+1}) - V(\xi_k) \leq -\ell(\xi_k, \mathbf{u}_k^*), \quad (21)$$

where $\ell(\cdot)$ denotes the stage cost and $\mathbf{u}_k^*$ is the optimal control input at time step k . Since the stage cost $\ell(x_k, u_k^*)$ is positive definite under the stated assumptions, the optimal value function serves as a Lyapunov function, implying local asymptotic stability in a neighborhood of the reference trajectory under the above conditions. $\square$

4. Experiment results and discussion

4.1. Background and track characteristics of the A2RL competition

The Abu Dhabi Autonomous Racing League (A2RL), initiated and organized by ASPIRE, is an international top-tier autonomous racing event designed to advance artificial intelligence and autonomous driving technologies through extreme competition. A2RL features three main formats: autonomous car racing, drone racing, and autonomous buggy racing. The car racing competition takes place at the Yas Marina Circuit in Abu Dhabi, as illustrated in Fig. 4. The circuit spans 5268 m and features 16 corners, consisting of 9 left and 7 right turns. Specifically, corners T5, T6, T9, and T13 are selected as representative examples to characterize track features, while lettered markers (A, B, C, E, F) indicate the local environmental conditions surrounding. It incorporates a combination of technical turns, long straights, and sections with varying curvature. In particular, the ST sector, characterized by densely distributed corners and pronounced curvature variations, presents a highly challenging environment for testing the dynamic performance of high-speed autonomous driving systems.

4.2. Experimental platform and configuration

To evaluate the control performance of the proposed algorithm under high-speed conditions, this study employs the high-fidelity autonomous driving simulation platform AutoVerse provided by the competition organizers. Built on the ROS2 framework, the platform integrates high-precision vehicle dynamics modeling with multi-sensor simulation capabilities, including LiDAR, cameras, and IMU, enabling realistic reproduction of vehicle dynamics and the actual characteristics of the Yas Marina Circuit. The overall architecture of the autonomous racing system, illustrated in Fig. 5, consists of three functional modules and the vehicle physical system: the localization module for state

Table 1

Key simulation and implementation parameters.

Category	Description / Value
Vehicle mass	760 kg
Front axle distance (l_f)	1.75 m
Rear axle distance (l_r)	1.35 m
Tire radius	0.31 m
Maximum engine speed	7000 rpm
Total track length	5268 m
Maximum planned speed	70 m/s
Discretization method	First-order Euler discretization (Li et al., 2024)
Sampling time (Δt)	0.125 s
Prediction horizon (N)	20
Nonlinear solver	IPOPT
Initialization strategy	Warm start using previous optimal solution

estimation and environment perception, the planning module for generating reference trajectories and optimal paths, and the control module, which executes longitudinal and lateral control strategies to precisely regulate braking, throttle, steering, and gear shifting. The experimental platform employed an Intel Core i7-13650HX processor with an NVIDIA RTX 4060 discrete GPU running on Ubuntu 22.04.

To comprehensively assess the algorithm's performance under extreme conditions, two representative test scenarios were designed: (1) track conditions with frequent curvature variations; and (2) full-track test conditions.

Fig. 6 depicts the expected speed distribution of the autonomous racing vehicle along the trajectory on the test track, with speeds ranging from 0 to 70 m/s. The trajectory is color-coded from blue to red, representing low to high speeds. Higher speeds are observed on straight sections or segments with gentle curvature, whereas speeds decrease markedly in curves or regions with rapid curvature variation. This visualization highlights the speed adaptability requirements across different track sections, providing a basis for controller parameter tuning and strategy formulation.

To assess the control performance of the proposed CA-PF-MPCC algorithm under high-speed conditions with frequent curvature variations, four representative benchmark algorithms were selected for comparison: the Feedforward-Feedback control algorithm (Kapania & Gerdes, 2015), Model Predictive Control with Control Barrier Function(MPC-CBF) (Zeng et al., 2021), the standard Model Predictive Contouring Control algorithm(MPCC) (Liniger et al., 2015), and linear parameter varying-model predictive control(LPV-MPC) (Alcalá et al., 2020). In the following experiments, all algorithms are implemented using the same vehicle model, sampling time, solver configuration, and track representation, while all MPC-based controllers employ warm-start initialization from the previous optimal solution to ensure consistent initialization and a fair comparison. Controller-specific parameters, including feedback gains and cost weights, are tuned individually for best performance following established practices in the literature and preliminary experiments. The detailed simulation and implementation parameters are summarized in Table 1. For the proposed CA-PF-MPCC, the key parameters are configured as $\alpha_0 = 0.85$, $\alpha_1 = 25$, $\beta_1 = 8$, $w_{c,0} = 45$, and $w_{l,0} = 7$. Moreover, the evaluation metrics adopted in this study include the race time T_r , lateral tracking error E_l , yaw error E_y , and yaw rate R_y . Let the true vehicle state at the k th sampling instant be (x_k, y_k, φ_k) , and let the state of the reference trajectory at the closest projection point be $(x_{r,k}, y_{r,k}, \varphi_{r,k})$. The lateral error, yaw error, and yaw rate are defined as follows:

$$E_l = |(x_k - x_{r,k}) \sin \varphi_{r,k} + (y_k - y_{r,k}) \cos \varphi_{r,k}|, \quad (22)$$

$$E_y = |\text{wrap}(\varphi_k - \varphi_{r,k})|, \quad (23)$$

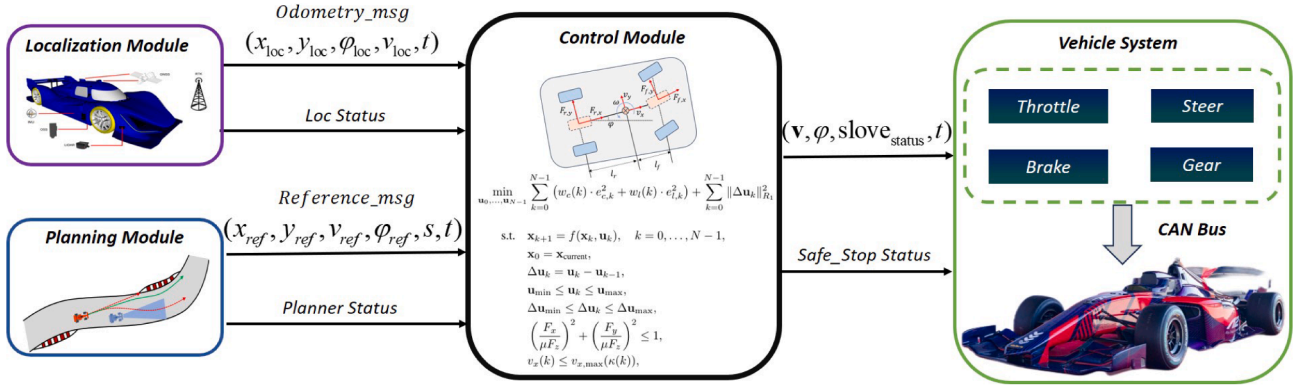


Fig. 5. Overall architecture of the autonomous racing system.

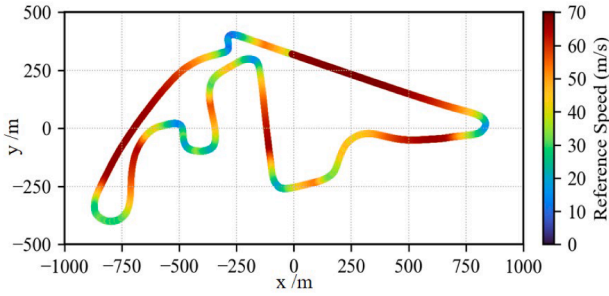


Fig. 6. Speed distribution along the vehicle's trajectory on the test track.

$$R_y = \left| \frac{\varphi_{k+1} - \varphi_k}{\Delta t} \right|. \quad (24)$$

where $\text{wrap}(\cdot)$ denotes an angle-wrapping function that confines angles to the interval $(-\pi, \pi]$, and Δt represents the control sampling period. To ensure experimental fairness and minimize the influence of random disturbances, all algorithms utilized a consistent perception, localization, and planning framework. Each algorithm was independently executed 50 times, with all relevant data accurately recorded. The mean values of each metric, namely $T_{m,r}$, $E_{m,l}$, $E_{m,y}$, and $R_{m,y}$, together with their corresponding standard deviations σ_{T_r} , σ_{l_e} , σ_{y_e} , and σ_{y_r} , were employed to comprehensively evaluate race efficiency, lateral tracking accuracy, yaw consistency, and overall vehicle attitude stability. Taking the lateral tracking error as an example, its mean value and standard deviation are computed as:

$$E_{m,l} = \frac{1}{N_n} \sum_{n=1}^{N_n} \left(\frac{1}{N} \sum_{k=1}^N E_l(k) \right), \quad (25)$$

$$\sigma_{l_e} = \sqrt{\frac{1}{N_n} \sum_{n=1}^{N_n} (E_{m,l} - \bar{E}_{m,l})^2}. \quad (26)$$

In the above expressions, N denotes the total number of control time steps in a single experimental run, N_n is the total number of independent runs, which is set to 50 in this study. $\bar{E}_{m,l}$ denotes the average of the mean lateral errors over all runs.

4.3. Track conditions with frequent curvature variations

This study systematically evaluated the control performance of the proposed CA-PF-MPCC algorithm under high-speed conditions with frequent curvature variations, with a particular emphasis on control stability and racing capability in complex scenarios. To capture the characteristics of such conditions, the ST section of the track-featuring dense

Table 2

Experimental results under conditions with frequently changing curvature.

	$T_{m,r}$, seconds σ_{T_r}	$E_{m,l}$, meters σ_{l_e}	$R_{m,y}$, rad/s σ_{y_r}	$E_{m,y}$, rad σ_{y_e}
Feedforward-Feedback	83.084 0.388	0.475 0.029	0.239 0.003	0.021 0.003
MPCC	82.536 0.634	0.316 0.021	0.247 0.007	0.034 0.005
MPC-CBF	84.110 0.136	0.503 0.038	0.316 0.005	0.026 0.005
LPV-MPC	81.541 0.225	0.323 0.017	0.215 0.003	0.023 0.004
CA-PF-MPCC	79.821 0.179	0.189 0.011	0.177 0.002	0.021 0.003

curves and pronounced curvature fluctuations—was selected as a representative test segment. This section measures approximately 3000 m in length, and the maximum planned speed of the autonomous race car was set to 70 m/s, thereby ensuring both the effectiveness and representativeness of the experimental design. The results obtained under curvature-varying conditions are summarized in Table 2.

In the table, the proposed CA-PF-MPCC algorithm outperforms the other compared algorithms in key metrics such as race time, tracking accuracy, and heading change rate, fully demonstrating its comprehensive control advantages in complex conditions with high speeds and frequently changing curvature. Specifically, the CA-PF-MPCC algorithm achieved reductions in race time of 3.263 s, 2.715 s, 4.289 s, and 1.720 s compared with Feedforward-Feedback, MPCC, MPC-CBF, and LPV-MPC, respectively, indicating a significant advantage in dynamic control aggressive. In terms of tracking accuracy, the algorithm achieved reductions in mean error of 0.286 m, 0.127 m, 0.314 m, and 0.134 m relative to the three baselines, thereby improving path following accuracy. With respect to yaw error, the CA-PF-MPCC algorithm achieved the same optimal level as Feedforward-Feedback, although the latter demonstrated slightly lower variance and marginally greater stability. With respect to yaw rate, the CA-PF-MPCC algorithm achieved reductions of 0.062 rad/s, 0.070 rad/s, 0.139 rad/s, and 0.038 rad/s compared with Feedforward-Feedback, MPCC, MPC-CBF, and LPV-MPC, respectively. These reductions effectively mitigated the “head-swaying” phenomenon observed when the autonomous race car negotiated high-speed corners or sections with large curvature, thereby enhancing vehicle attitude stability. Such improvements can be attributed to the curvature-aware adaptive error weighting mechanism integrated into CA-PF-MPCC, which allows real-time adjustment of error weights in response to curvature variations, balancing smooth vehicle posture transitions with racing performance under high-speed, changing curvature scenarios.

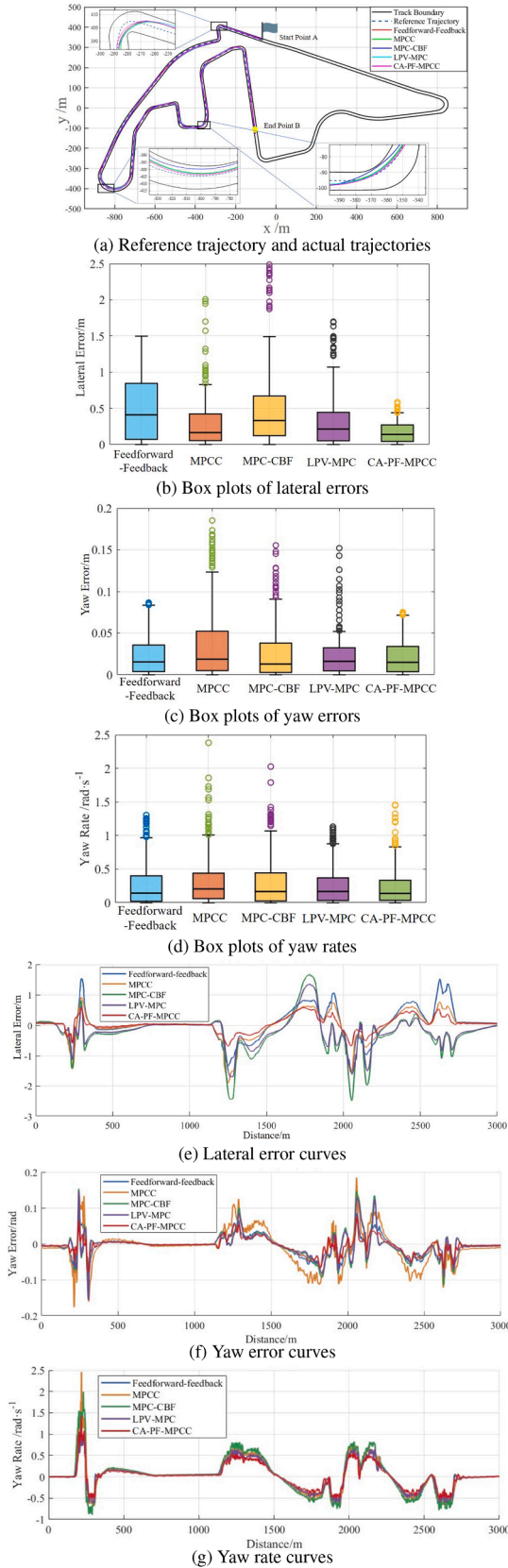


Fig. 7. Representative test results of different algorithms under conditions with frequent curvature variations.

Table 3

Experimental results under full-track test conditions.

	$T_{m,r}$, seconds σ_{rt}	$E_{m,l}$, meters σ_{le}	$R_{m,y}$, rad/s σ_{yr}	$E_{m,y}$, rad σ_{ye}
Feedforward-Feedback	132.595 0.548	0.421 0.009	0.209 0.003	0.018 0.001
MPCC	131.844 0.793	0.273 0.018	0.221 0.017	0.026 0.003
MPC-CBF	133.217 0.349	0.452 0.015	0.247 0.003	0.017 0.003
LPV-MPC	130.342 0.453	0.291 0.015	0.185 0.003	0.015 0.002
CA-PF-MPCC	128.491 0.393	0.101 0.008	0.159 0.003	0.018 0.002

Fig. 7 presents representative test results of five algorithms on the ST section of the track, including actual driving trajectories, box plots, and corresponding distance-based curves of lateral error, yaw error, and yaw rate. The results indicate that the proposed CA-PF-MPCC algorithm consistently follows the reference trajectory across the ST section, with smaller deviations in curves with sharp curvature variations. The lateral error box plots further reveal that the errors of the Feedforward-Feedback and MPC-CBF algorithms are more dispersed, and both MPCC and MPC-CBF exhibit multiple error points exceeding 0.2 m.

In contrast, CA-PF-MPCC achieves the lowest median, the most concentrated distribution, and the smallest peak error, demonstrating a clear advantage. These findings suggest that the algorithm effectively suppresses lateral deviations in track segments with frequent curvature changes. The performance improvement primarily stems from the introduction of a curvature-aware adaptive error weighting mechanism, which adjusts longitudinal and lateral error weights in real time according to track curvature, thereby enhancing tracking accuracy in high-speed cornering. In addition, the integration of tire force constraints via the friction ellipse and the curvature-speed coupling constraint effectively mitigates skidding and sliding under aggressive maneuvers. With respect to heading control, CA-PF-MPCC maintains a median yaw rate below 0.15 rad/s with relatively small fluctuations, indicating stable and well-regulated vehicle dynamics. Although the LPV-MPC algorithm exhibits fewer outliers in yaw rate, its median yaw rate remains higher than that of CA-PF-MPCC. Meanwhile, CA-PF-MPCC also achieves a more concentrated yaw error distribution with fewer outliers, reflecting superior heading regulation performance. Overall, these results highlight the superior stability and pose consistency of the proposed algorithm under high-speed and frequent curvature variations, ultimately improving racing performance.

4.4. Full-track test conditions

Based on the tests under frequently varying curvature conditions, full-track experiments were conducted to further evaluate the performance of the proposed CA-PF-MPCC algorithm under high-speed conditions. The circuit has a total length of 5,268 m, with a maximum planned speed of 70 m/s. To ensure experimental fairness, each algorithm was independently executed 50 times, and all relevant data were precisely recorded. Key performance indicators—including lap time, lateral tracking error, yaw error, and yaw rate—were statistically analyzed to comprehensively assess the algorithms across the entire track. The results are summarized in Table 3.

As shown in the table, the CA-PF-MPCC algorithm demonstrates significant advantages in racing performance, lateral tracking accuracy, and attitude stability under high-speed, full-track conditions. Specifically, CA-PF-MPCC achieves an average lap time of 128.491 s, which is 4.104 s, 3.353 s, 4.726 s, and 1.851 s shorter than those obtained by the Feedforward-Feedback, MPCC, MPC-CBF, and LPV-MPC algorithms,

Table 4

Ablation study results for different variants of the CA-PF-MPCC framework.

	$T_{m,r}$, seconds σ_{r1}	$E_{m,l}$, meters σ_{le}	$R_{m,y}$, rad/s σ_{yr}	$E_{m,y}$, rad σ_{ye}
MPCC	82.536 0.634	0.316 0.021	0.247 0.007	0.034 0.005
PF-MPCC	81.712 0.452	0.240 0.015	0.225 0.004	0.028 0.004
CA-MPCC	80.415 0.220	0.203 0.012	0.178 0.003	0.022 0.003
CA-PF-MPCC	79.821 0.179	0.189 0.011	0.177 0.002	0.021 0.003

respectively, indicating a substantial improvement in overall racing performance. In terms of lateral tracking accuracy, CA-PF-MPCC reduces the mean lateral error by 0.320 m, 0.172 m, 0.351 m, and 0.190 m compared with the baseline algorithms, demonstrating its effectiveness in suppressing lateral deviations on complex tracks. Regarding attitude control, the yaw error of CA-PF-MPCC is slightly higher than that of MPC-CBF and LPV-MPC. Nevertheless, it achieves the lowest yaw rate among all evaluated algorithms, reflecting excellent heading control stability. This trade-off suggests that CA-PF-MPCC prioritizes dynamic stability and smooth heading regulation over aggressive instantaneous yaw correction, which is particularly desirable in high-speed racing scenarios.

Fig. 8 presents a representative full-track test of the five algorithms, including the actual driving trajectories, box plots, and corresponding distance-based curves of lateral error, yaw error, and yaw rate. The results in Fig. 8(a) indicate that CA-PF-MPCC closely follows the reference trajectory throughout the entire track, with minimal deviation in high-curvature corners, demonstrating superior curvature adaptability and path-keeping capability. Statistical analysis of the lateral error shows that the median for CA-PF-MPCC is only 0.09 m, with the smallest interquartile range and the most concentrated distribution, outperforming the other four algorithms and significantly enhancing tracking accuracy and stability under complex conditions. In terms of attitude control, the yaw error distributions of the five algorithms are generally comparable. Although the LPV-MPC algorithm exhibits a slightly more concentrated yaw error distribution, it also presents several large outliers. By contrast, the maximum yaw errors of CA-PF-MPCC and the Feedforward-Feedback algorithm remain below 0.1 rad, indicating more reliable heading regulation. Besides, CA-PF-MPCC exhibits the lowest yaw rate and smallest variance among all algorithms, highlighting its advantage in suppressing high-frequency attitude adjustments and maintaining smooth vehicle operation.

4.5. Ablation study

To quantitatively evaluate the impact of the two key components of the proposed CA-PF-MPCC algorithm, namely curvature aware adaptive error weighting mechanism and physical feasibility constraints, this paper conducted a systematic ablation study under track conditions with frequent curvature variations. Four controller variants were tested: (1) the baseline MPCC with fixed contouring and lag error weights and without feasibility constraints; (2) PF-MPCC, which incorporates physical feasibility constraints including tire force and curvature-speed limits but uses fixed error weights; (3) CA-MPCC, which employs curvature aware adaptive weighting but no physical feasibility constraints; and (4) the CA-PF-MPCC algorithm, which integrates both curvature aware adaptive weighting and physical feasibility constraints. Each variant was evaluated over 50 laps, and the ablation results are summarized in Table 4

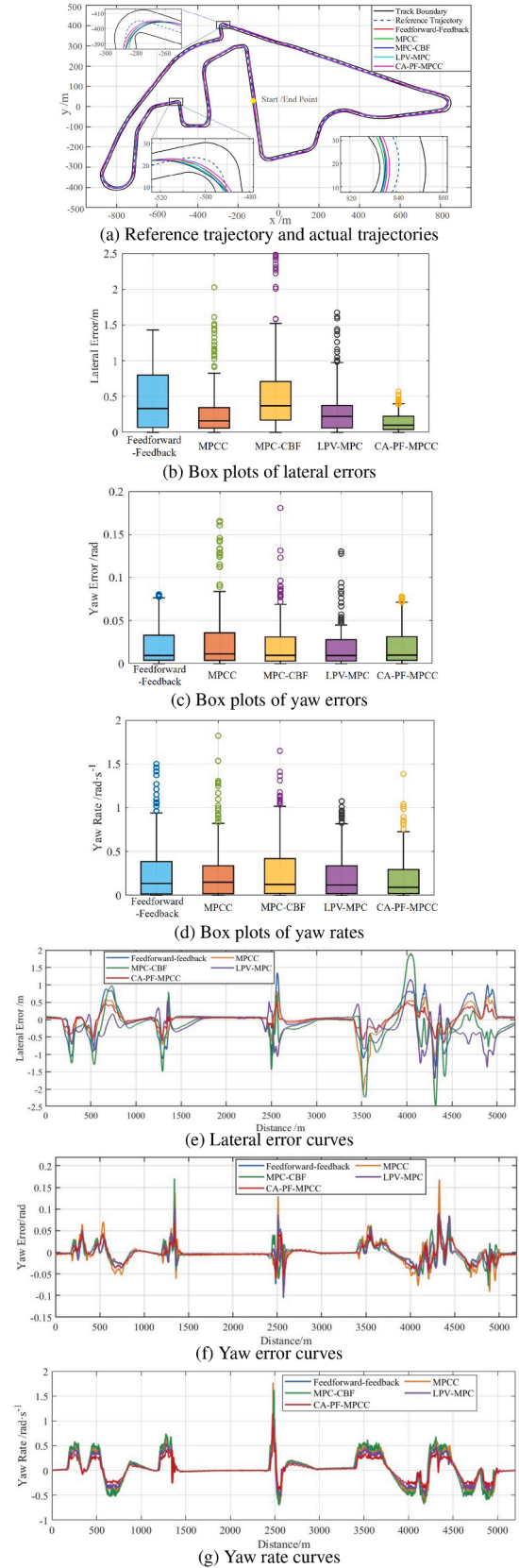


Fig. 8. Representative test results of different algorithms under full-track conditions.

Table 5
Comparison of different curvature-dependent weighting forms.

Method	$T_{m,r}$ (s)	$E_{m,l}$ (m)	$R_{m,y}$ (rad/s)	$E_{m,y}$ (rad)
MPCC-Fixed	82.634	0.301	0.250	0.033
MPCC-Linear	83.374	0.347	0.286	0.039
CA-PF-MPCC	79.681	0.183	0.174	0.020

From the table, it can be seen that PF-MPCC, by introducing only physical feasibility constraints, reduces the lap time to 81.712 s, representing a 1.0% improvement over the baseline MPCC. Meanwhile, the mean yaw rate and yaw error decrease to 0.225 rad/s and 0.028 rad, achieving reductions of 8.9% and 17.6%, respectively. indicating enhanced stability and reduced sensitivity to aggressive maneuvers in high curvature regions. The lateral error decreases moderately from 0.316 m to 0.240 m, suggesting that physical feasibility primarily enhances stability rather than tracking accuracy.

Besides, by incorporating curvature aware adaptive error weighting mechanism, CA-MPCC delivers substantially greater performance gains. The average lap time is reduced to 80.415 s, representing a 2.6% improvement relative to MPCC. The mean lateral error decreases markedly to 0.203 m, corresponding to a 35.8% reduction, while the mean yaw rate and yaw error are further reduced to 0.178 rad/s and 0.022 rad. These results demonstrate that dynamically adjusting the relative importance of contouring and lag errors according to local track curvature effectively improves tracking accuracy and overall racing efficiency, especially under frequent curvature transitions. than path tracking accuracy.

The CA-PF-MPCC delivers the best overall performance. These results confirm that physical feasibility constraints enhance stability, while the curvature-aware adaptive weighting mechanism improves tracking accuracy and lap time. Their combination provides complementary benefits, yielding superior overall performance.

4.6. Comparison of different curvature-dependent weighting forms

To examine whether the observed performance gains are specific to the sigmoid formulation, this paper performs a comparison of different curvature-dependent weighting forms, each executed 10 times, including three representative strategies: MPCC-Fixed, which uses constant weights; MPCC-Linear, which applies linearly varying curvature-dependent weights; and CA-PF-MPCC, which employs bounded sigmoid-based curvature-aware weighting. The quantitative results are summarized in Table 5.

As shown in the table, CA-PF-MPCC with bounded sigmoid - based weighting outperforms both MPCC with fixed weights and MPCC with linear curvature - dependent weighting. Specifically, CA-PF-MPCC achieves an average lap time of 79.681 s, which is 2.953 s and 3.993 s faster than the other two methods, respectively. In terms of tracking accuracy, the mean lateral error is reduced by 0.118 m and 0.164 m compared with MPCC - Fixed and MPCC - Linear, respectively. In addition, both the mean yaw rate and mean yaw error of CA-PF-MPCC are lower than those of the other weighting forms. These results indicate that the bounded sigmoid mapping not only improves lap - time efficiency but also substantially enhances lateral and heading tracking performance.

4.7. Sensitivity analysis of hyperparameters

To evaluate the robustness of the curvature-aware weighting mechanism and to assess the sensitivity of the algorithm to different parameters, this paper conducts a systematic sensitivity analysis on the key parameters of the mechanism, including the sigmoid slope α_0 , the contouring error weight gain α_1 , and the lag error weight gain β_1 . The results of the sensitivity experiments for each parameter are summarized in Table 6.

Table 6
Sensitivity analysis results of key parameters.

Parameters	Variation	$T_{m,r}$ (s)	$E_{m,l}$ (m)	$R_{m,y}$ (rad/s)	$E_{m,y}$ (rad)
Sigmoid slope α_0	$0.5\alpha_0$	81.825	0.254	0.208	0.025
	α_0	79.834	0.192	0.179	0.021
	$1.5\alpha_0$	80.351	0.223	0.232	0.029
Weight gain α_1	$0.5\alpha_1$	80.582	0.213	0.191	0.019
	α_1	79.815	0.188	0.174	0.020
	$1.5\alpha_1$	79.967	0.172	0.166	0.021
Weight gain β_1	$0.5\beta_1$	80.420	0.183	0.176	0.019
	β_1	79.754	0.181	0.173	0.021
	$1.5\beta_1$	79.479	0.247	0.181	0.023

Table 7
Results of six experimental runs on the Suzuka Circuit.

Run Number	T_r (s)	E_l (m)	R_y (rad/s)	E_y (rad)
1	117.49	0.236	0.173	0.020
2	117.96	0.248	0.175	0.021
3	118.04	0.257	0.174	0.021
4	117.38	0.234	0.173	0.020
5	117.91	0.241	0.174	0.021
6	118.12	0.259	0.175	0.021

From the table, it can be seen that the parameter α_0 determines the steepness of the curvature-dependent sigmoid function and thus directly affects how rapidly the tracking error weights are adjusted in response to curvature variations. When α_0 is reduced to $0.5\alpha_0$, the weight adaptation becomes overly smooth, leading to insufficient penalization in high - curvature segments and consequently larger lateral deviation and longer lap times. Conversely, increasing α_0 to $1.5\alpha_0$ results in overly aggressive weight transitions, slightly degrading yaw stability, as indicated by the increased yaw rate and yaw error. The parameter α_1 primarily scales the lateral tracking error weight. Increasing α_1 from $0.5\alpha_1$ to $1.5\alpha_1$ produces a monotonic reduction of the average lateral error from 0.213 m to 0.172 m, confirming its dominant influence on lateral path adherence. Moreover, the parameter β_1 mainly affects the vehicle's progression efficiency. As β_1 varies from $0.5\beta_1$ to $1.5\beta_1$, the lap time decreases from 80.421 s to 79.379 s, while the mean lateral error increases by 0.044 m; thus, increasing β_1 improves racing efficiency but also influences the vehicle's yaw behavior. Overall, the results indicate that the proposed curvature - aware adaptive weighting mechanism exhibits a certain degree of robustness to parameter variations.

4.8. Algorithm evaluation on the suzuka circuit

To evaluate the robustness of the proposed algorithm across different racing circuits, experiments were conducted on the Suzuka Circuit, which has a total length of 5.793 km and features 18 corners arranged in a distinctive figure-eight layout. The track exhibits diverse geometric characteristics, including high-speed S-curves, tight hairpins, and long straight sections, posing significant challenges for high-performance trajectory tracking. On this circuit, six independent experimental runs were carried out under identical conditions, and the corresponding quantitative results are summarized in Table 7.

Fig. 9 illustrates the vehicle state evolution during the fifth run at several representative locations along the track. Across the six runs, the proposed controller achieves an average lap time of 117.82 s with a variation of less than 1 s, indicating strong repeatability and consistency. Furthermore, the mean lateral tracking error remains bounded at approximately 0.246 m, while the mean yaw error is maintained around 0.0206 rad. These performance metrics exhibit smooth variations across repeated experiments, reflecting stable closed-loop behavior and robustness of the proposed algorithm under complex track geometries.

Table 8
Qualitative comparison of curvature-related MPCC strategies.

Method	Curvature utilization	Level	Learning-based	Main objective
Ci-MPCC	Mapped to reference speed profile	Trajectory planning	No	Optimize velocity distribution along the planned trajectory
RLA-MPCC	Implicitly learned through policy or value function	Control	Yes	Improve performance via data-driven adaptation
CA-PF-MPCC	Explicit curvature-dependent adaptive weighting	Control	No	Dynamically balance contouring and lag errors for high-speed racing

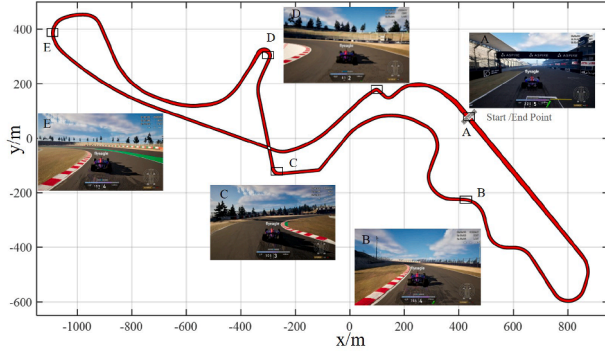


Fig. 9. Vehicle state sequence of the autonomous race car during the fifth experiment.

4.9. Qualitative analysis of the curvature-adaptive strategy

To clarify the distinction between the proposed curvature-adaptive strategy and existing curvature-related MPCC approaches, this paper performs a qualitative comparison with Curvature-Integrated MPCC (Ci-MPCC) (Liniger et al., 2015) and Reinforcement-Learning-Assisted MPCC (RLA-MPCC) (Evans et al., 2023), summarized in Table 8.

From the table, Ci-MPCC improves racing performance by mapping the curvature of the track centerline to a reference velocity through a predefined mapping function, dynamically planning a velocity that adapts to the track curvature. Moreover, this method is applied at the trajectory planning level. In this formulation, curvature information is first mapped to a reference velocity distribution, which is then incorporated into the optimization objective function to plan the velocity profile along the preplanned local trajectory. In contrast, the proposed curvature-aware adaptive error weighting mechanism is embedded into the objective function through a state-dependent adaptive weighting scheme. This scheme dynamically balances the contouring error and lag error over the prediction horizon, thereby achieving an explicit trade-off between tracking accuracy and stability under rapidly varying curvature conditions, which ultimately enhances racing performance. In addition, this mechanism operates at the control level. From a mathematical standpoint, the curvature-aware adaptive error weighting mechanism dynamically adjusts the weighting coefficients in the objective function, and adaptively regulates the relative importance of contouring error and lag error within the prediction horizon, enabling a balance between performance and stability under frequent curvature variations to improve racing capability.

Furthermore, compared with RLA-MPCC, which relies on a learning paradigm whose performance depends on the training environment, data distribution, and offline learning process, the proposed method adopts a model-based and optimization-driven design. By avoiding learning components, the proposed curvature-aware adaptive error weighting mechanism is less sensitive to data quality and environmental variations, and provides a fully deterministic and interpretable formulation with explicitly defined bounds on vehicle behavior. Importantly, no training data are required.

In summary, the CA-PF-MPCC algorithm demonstrates clear superiority in both racing performance and control stability compared with the other methods, fully validating its adaptability and practical appli-

cability under high-speed, complex track conditions, and providing a robust and efficient solution for autonomous racing in challenging environments.

5. Conclusions

At high speeds, especially on tracks with frequently changing curvature, the stability of vehicle control systems tends to deteriorate, thereby constraining both the racing performance and operational safety of autonomous racing cars. To address this challenge, an enhanced model predictive contouring control algorithm, referred to as CA-PF-MPCC, is developed. Within the MPCC framework, a curvature-based dynamic error adaptation mechanism is introduced, in which a dynamic weighting function adaptively adjusts the penalties of contouring and lag errors, enhancing the controller's responsiveness to track sections with diverse geometric characteristics. Furthermore, physically feasible constraints are incorporated, including tire force limitations based on the friction ellipse model and velocity-curvature coupling constraints. These constraints effectively suppress physical instabilities during near-limit maneuvers, ensuring system stability and safety at high speeds. Experimental results under conditions of frequent curvature variations demonstrate that, compared with baseline methods, the proposed algorithm reduces racing times by 3.263 s, 2.715 s, 4.289 s, and 1.720 s, respectively, while significantly mitigating heading oscillations, thus improving vehicle attitude stability. Furthermore, the CA-PF-MPCC demonstrates real-time capability, achieving an average computation time of 0.012 s per control cycle, highlighting its practical applicability for high-speed autonomous racing. These results underscore the algorithm's potential for high-speed autonomous racing applications.

Future work will focus on further enhancing the performance and robustness of the proposed autonomous racing framework. In particular, the algorithm will be validated in real-world environments, considering the impact of sensor noise and actuation delay on performance. Furthermore, the integration of reinforcement learning techniques will be investigated to support end-to-end autonomous control, thereby improving the adaptability and decision-making intelligence of autonomous racing vehicles in complex and dynamic environments. In parallel, more accurate vehicle dynamics models will be employed to better capture real vehicle behavior.

CRedit authorship contribution statement

Shengshan Ma: Writing – original draft, Visualization, Validation, Methodology, Formal analysis; **Kaiyan Zhao:** Writing – original draft, Software, Investigation; **Ren Jin:** Writing – review & editing, Visualization; **Shoukun Wang:** Writing – review & editing, Funding acquisition; **Junzheng Wang:** Writing – review & editing, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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